

AJINKYA PRASHANT GOKHALE

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Education

University of Stuttgart

MSc. Infotech, Major: Computer Hardware/Software Engineering

Oct. 2023 – Present

Stuttgart, Germany

Veermata Jijabai Technological Institute

Bachelor of Technology in Electronics and Telecommunications Engineering (GPA: 8.32/10)

July 2020 – May 2023

Mumbai, India

Government Polytechnic Amravati

Diploma in Electronics and Telecommunications Engineering (96.68%)

July 2017 – May 2020

Amravati, India

Experience

Nineti GmbH (Startup)

Working Student (Former Intern)

Feb. 2024 – Present

Stuttgart, Germany

- **Founding Engineer:** Spearheaded development of Stromleser IoT devices, mobile application, and **AWS backend infrastructure**, contributing to the company's foundational technology stack.
- **AWS Cloud Solutions:** Implemented scalable cloud infrastructure managing data from **10,000+ IoT devices** using EC2, Lambda, DynamoDB, and IoT Core, ensuring 99.9% uptime and real-time data processing.
- **Hardware Design:** Designed and tested **20+ electronic circuits** using KiCad, demonstrating expertise in PCB design, component selection, and hardware validation processes.
- **Firmware Optimization:** Optimized embedded firmware resulting in **25% reduction** in response time and improved power efficiency for battery-operated devices.

Nomadists India Pvt. Ltd.

Full Stack Developer Intern

May 2020 – Aug. 2020

Pune, India

- Developed responsive web applications using **HTML5, CSS3, JavaScript**, focusing on cross-browser compatibility and mobile-first design principles.
- Implemented database solutions using **SQL and ORM frameworks**, optimizing query performance and ensuring data integrity across multiple platforms.
- Collaborated using **Git version control** and agile methodologies, maintaining code quality through peer reviews and continuous integration practices.

Projects

SwapMails - AWS Serverless Email Platform | Full-Stack Web App and Chrome Extension

Jan. 2025 – Present

- Architected **event-driven serverless architecture** using AWS Lambda, SES, and DynamoDB with **Infrastructure as Code** (CDK v2), serving 1000+ users with 99.9% uptime and automatic scaling.
- Built **GDPR-compliant** email processing system with automated DynamoDB TTL cleanup, Lambda-triggered parsing, and SES integration achieving **sub-60 second** delivery times.
- Developed **Chrome extension** published on Chrome Web Store with 500+ active users, featuring one-click email generation, real-time inbox synchronization, and seamless integration with 100+ websites.
- Engineered React.js frontend with responsive design supporting real-time email updates, custom domain integration, and multi-language support, deployed via AWS CloudFront CDN for global performance optimization.
- Implemented **RESTful API architecture** using AWS API Gateway with custom authentication, rate limiting, and comprehensive error handling, serving 10,000+ API calls daily.
- Designed privacy-by-design architecture with automatic data deletion, AES-256 encryption, comprehensive audit logging, and zero-knowledge data processing demonstrating advanced cloud security practices.
- Optimized cost efficiency achieving **95% cost reduction** compared to traditional hosting through serverless architecture, with automated scaling handling traffic spikes up to 1000x baseline load.

Gesture Recognition using Time-of-Flight Camera | Python, OpenCV, Computer Vision

Jan. 2023

- Developed hand gesture recognition system using Python and Time-of-Flight camera achieving **95% accuracy** in real-time gesture detection and classification.
- Implemented **computer vision algorithms** for gesture interpretation, resulting in **20% reduction in false positives** through advanced filtering and machine learning techniques.
- Enhanced user interface experience by integrating gesture control technology, demonstrating practical application of depth sensing and motion tracking algorithms.

Technical Skills

Cloud Platforms: AWS (Expert) - Lambda, EC2, S3, DynamoDB, RDS, CloudFormation, CloudWatch, IAM, VPC, API Gateway, SES, IoT Core — Google Cloud Platform

Programming Languages: Python, Java, JavaScript, C/C++, Rust, HTML5/CSS3, TypeScript

Backend & APIs: Node.js, Express.js, RESTful APIs, Microservices Architecture, Serverless Computing, Event-Driven Architecture

Frontend Development: React.js, HTML5, CSS3, JavaScript ES6+, Chrome Extension Development, Responsive Design

Infrastructure as Code: AWS CDK v2, CloudFormation, Terraform — CI/CD Pipelines - GitHub Actions, AWS CodePipeline, Jenkins

Databases: Amazon DynamoDB, Amazon RDS (PostgreSQL, MySQL), SQL, NoSQL, Database Design, Data Modeling

DevOps & Monitoring: Docker, AWS CloudWatch, X-Ray, Linux Administration, Performance Monitoring, Log Analysis

IoT & Embedded Systems: AWS IoT Core, MQTT, Raspberry Pi, Arduino, ESP32/NodeMCU, Sensor Integration, Firmware Development

Development Tools: VS Code, Git, GitHub, AWS CLI, Postman, Jira, KiCad, Fusion 360

Security & Compliance: GDPR Compliance, AWS IAM, SSL/TLS, Network Security, Data Privacy, Encryption

Languages

English: Professional Proficiency

German: Intermediate (B1 Level)

Hindi: Native Speaker

Hobbies

Local LLMs: Optimizing inference, testing quantized models, and exploring open-source AI on personal hardware.

Self-Hosting: Running services like Home Assistant and Nextcloud on a personal Kubernetes cluster.

Homelab: Tinkering with container orchestration, networking, and infrastructure automation.

Extracurricular Activities

GPHReviews - Tech Content Creator

Spring 2017 – Oct. 2023

Digital Content Creator & Community Builder

YouTube Platform

- Built and engaged a community of **42,000+** subscribers through consistent tech review content, demonstrating digital marketing and audience engagement skills.
- Established partnerships with **iOS application developers** and tech companies, facilitating product launches and creating mutually beneficial promotional campaigns.
- Created an active online community of Apple technology enthusiasts, fostering discussions on emerging technologies and product innovations.